

CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter reviews the literature, technical principles, and existing systems that form the foundation of CacaoScan. Manual post-harvest sorting in Zamboanga City cannot reliably segregate individual cacao beans by the three Philippine cultivar groups (Criollo, Forastero, and Trinitario) while simultaneously grading surface quality under PNS/BAFS 58:2019 at a cost accessible to barangay-level cooperatives.

The review integrates twenty verified related studies and industry sources on agricultural computer vision, YOLO-based defect detection, IoT actuated sorting, and Philippine cacao standards published from 2020 to 2026, supplemented by foundational edge computing and deep learning references where required for theoretical grounding. The discussion is organized into Related Theories, which explain the principles behind dual-model YOLOv8 inference, IoT actuation, and Philippine cacao grading standards, and Related Literature and Systems, which examine twenty published works that validate individual components of CacaoScan while showing that no existing system combines all proposed features in one machine.

The first four related studies include supporting figures. The chapter concludes with Table 2, a summary, and a synthesis linked to the project objectives: Model A variety classification, Model B quality grading into Export-Grade, Needs Drying, and Rejected bins, ESP32-controlled physical sorting, and React Native batch monitoring with digital certification logs.

This chapter supports the CacaoScan prototype architecture, which combines cloud-hosted dual YOLOv8 inference with conveyor-fed batch staging, an enclosed vision tower, ESP32-S3 multi-actuator discharge, and React Native plus web dashboard monitoring for standalone cooperative deployment in Zamboanga City. Model B quality grading in the proposed system is further interpreted using the visual defect framework established by PNS/BAFS 58:2019 for Philippine cacao beans.

2.1 Related Theories

2.1.1 Computer Vision and Deep Learning for Agricultural Quality Assessment

Computer vision is a subfield of artificial intelligence that enables machines to interpret and understand visual information from digital images and video streams. In agricultural post-harvest processing, computer vision replaces subjective human visual inspection with algorithmic analysis of color, texture, shape, and surface defect patterns on individual commodities (Szeliski, 2022). Deep learning architectures, particularly convolutional neural networks, have become the dominant approach for agricultural image classification because they learn hierarchical feature representations directly from labeled training data without requiring hand-crafted feature engineering (LeCun et al., 2015).

For cacao and other high-value crops, computer vision supports automated grading by detecting mold colonies, slaty coloration, shrinkage, and physical breakage on bean surfaces, which are the same defect classes defined for CacaoScan and aligned with PNS/BAFS 58:2019. CacaoScan applies this theory by capturing high-resolution images of individual beans and routing them through Model A (variety) and Model B (quality) as defined in the project proposal.

2.1.2 YOLO Object Detection Architecture

You Only Look Once (YOLO) is a family of real-time object detection architectures that perform localization and classification in a single forward pass through a neural network, making them suitable for high-throughput sorting applications where latency must remain under two seconds per item. YOLOv8, developed by Ultralytics, introduces an anchor-free detection head, improved feature pyramid design, and optimized training pipelines that achieve state-of-the-art mean average precision on commodity hardware with CUDA acceleration (Jocher et al., 2023).

Recent post-harvest studies demonstrate that YOLOv8n variants can classify agricultural defects with mAP50 values above 0.79 and inference times as low as one millisecond per image on GPU-equipped workstations, confirming suitability for real-time automated sorting pipelines (Cordeiro et al., 2025). CacaoScan extends this theory by deploying two independently trained YOLOv8 models: Model A for variety classification of the three Philippine cultivar groups (Criollo, Forastero, and Trinitario) and Model B for quality grading on the same input image, enabling specialized accuracy in both dimensions without the performance trade-offs of a single multi-class generalist model.

CacaoScan adopts a dual-model architecture rather than a single multi-class YOLOv8 model because variety classification and PNS/BAFS 58:2019 quality grading represent distinct decision tasks with different label spaces, error costs, and dataset requirements. A single six-class or nine-class model would force shared feature optimization across cultivar morphology and surface defect detection, increasing confusion between visually similar Trinitario and Forastero beans and between borderline Needs Drying and Rejected defects. Independent Model A and Model B training allows specialized augmentation policies, class-balanced sampling, and per-task hyperparameter tuning, while decision fusion logic combines both outputs to determine the final physical bin assignment. Separate evaluation with distinct confusion matrices, precision, recall, and F1-score for each model satisfies the methodological requirement to report variety and quality performance independently rather than as a single aggregated accuracy figure.

2.1.3 Internet of Things and Edge Inference

The Internet of Things (IoT) refers to networks of physical devices embedded with sensors, actuators, and communication hardware that collect and exchange data over local or wide-area networks (Atzori et al., 2010). When combined with artificial intelligence inference at or near the data source, IoT systems form AIoT architectures that execute decision-to-action pipelines without continuous cloud dependency. Edge computing theory further supports local

inference by processing data on-premises, reducing latency and enabling operation in facilities with intermittent internet connectivity (Shi et al., 2016).

In automated sorting systems, IoT microcontrollers such as the ESP32 receive classification results from an inference server and translate them into physical actuator commands that divert items into designated output bins. CacaoScan implements this theory through an ESP32-controlled servo sorting arm that receives YOLOv8 inference outputs via local Wi-Fi and activates MG996R servo motors within a target latency of under two seconds per bean, completing the full sense-decide-act loop at the edge of the network.

2.1.4 Post-Harvest Cacao Grading Standards

Post-harvest cacao grading in the Philippines is governed by PNS/BAFS 58:2019, which specifies grading and classification requirements for cacao beans derived from harvested ripe pods of *Theobroma cacao* L. The standard defines quality thresholds for moisture content, mold presence, slaty beans, insect damage, and foreign matter, and establishes grading categories based on the proportion of defective beans determined through defined test methods aligned with ISO 2451:2017 (Philippine Bureau of Agriculture and Fisheries Standards, 2019).

At the variety level, the Philippine cacao industry recognizes three major cultivar groups cultivated nationwide: Criollo, Forastero, and Trinitario. Criollo is prized for fine flavor but is less common; Forastero is the most widely grown and hardy group; and Trinitario is a hybrid combining the productivity of Forastero with the flavor characteristics of Criollo (Department of Agriculture and Department of Trade and Industry, 2017). Accurate segregation of these three groups at post-harvest is essential because each carries distinct market valuation, yet cooperative sorting in Zamboanga City still relies on subjective visual judgment.

Export-grade beans must meet strict visual inspection criteria before entering international markets, making accurate defect detection at the cooperative level a critical control point in the cacao value chain. CacaoScan operationalizes this standard by training Model B to detect surface defects, including mold, slaty coloration, shrinkage, and breakage as listed in the project proposal, and training Model A to classify individual beans as Criollo, Forastero, or Trinitario, outputting three physical sorting bins: Export-Grade, Needs Drying, and Rejected.

2.1.5 Automated Sorting and AIoT Integration

Automated agricultural sorting integrates sensor input, classification logic, and mechanical actuation into a single closed-loop system that replaces manual hand-sorting with consistent, high-throughput processing. Theoretical models of human visual inspection further demonstrate that prolonged repetitive inspection tasks exhibit vigilance decrement, with accuracy declining as sorting sessions extend beyond continuous operation periods (Sturman et al., 2021). Automated systems eliminate this fatigue-related variability by applying uniform classification criteria to every item regardless of batch size or session duration.

AIoT sorting machines occupy a distinct design space between screen-only assessment tools, which detect quality visually but leave physical separation to human workers, and industrial-grade sorters priced beyond the reach of smallholder cooperatives. CacaoScan is positioned within this space as an affordable AIoT machine that combines dual-model computer vision, IoT actuation, mobile monitoring, and digital quality certification in a single system designed for barangay-level cacao processing facilities in the Zamboanga Peninsula.

Recent cacao and post-harvest studies from 2020 to 2026 reinforce the feasibility of computer-vision-based grading while highlighting the persistent actuation gap CacaoScan addresses. Opoku Eric et al. (2023) demonstrated enhanced feature extraction and neural network classification of cocoa beans with strong accuracy on quality categories but without Philippine cultivar classes or physical sorting integration. Cordeiro et al. (2025) validated YOLOv8 for multi-defect post-harvest detection at millisecond inference speeds on GPU hardware, supporting the Model B design rationale. Together with Zhinin Vera et al. (2023) and Nebrida (2025), the literature confirms that detection, classification, and physical sorting have been proven separately but not yet integrated in one cooperative-scale cacao system for the Zamboanga context.

2.2 Related Literature and Systems

2.2.1 Overview of Existing Cacao and Agricultural Sorting Systems

Existing agricultural quality assessment and sorting systems span three general categories: deep learning screen-based classifiers that detect defects visually but do not sort physically, IoT-integrated sorters that use simple sensors for physical separation without AI-based defect analysis, and industrial machines that combine both capabilities at costs inaccessible to smallholder cooperatives. The following twenty related studies represent the most directly comparable literature to CacaoScan, organized from cacao-specific and Philippine-context works to broader post-harvest AI, IoT sorting, and review sources published from 2020 to 2026.

Related Study 1: YOLOv8 Postharvest Defect Detection (Cordeiro et al., 2025)

Cordeiro, de Alencar Nääs, and Okano (2025) published a study on smart postharvest management of strawberries using YOLOv8-driven detection of defects, diseases, and maturity. The researchers compiled a custom dataset of 5,663 annotated strawberry images covering eight quality categories including anthracnose, gray mold, powdery mildew, uneven ripening, and physical defects, with data augmentation applied to improve generalization.

The YOLOv8n model trained over 200 epochs achieved a mAP50 of 0.79 with an inference time of one millisecond per image on GPU hardware, demonstrating suitability for real-time post-harvest quality classification. The study validates YOLOv8 as a viable architecture for automated agricultural defect detection at sorting speeds, though it addressed detection only without physical actuation, variety classification, or Philippine cacao context.

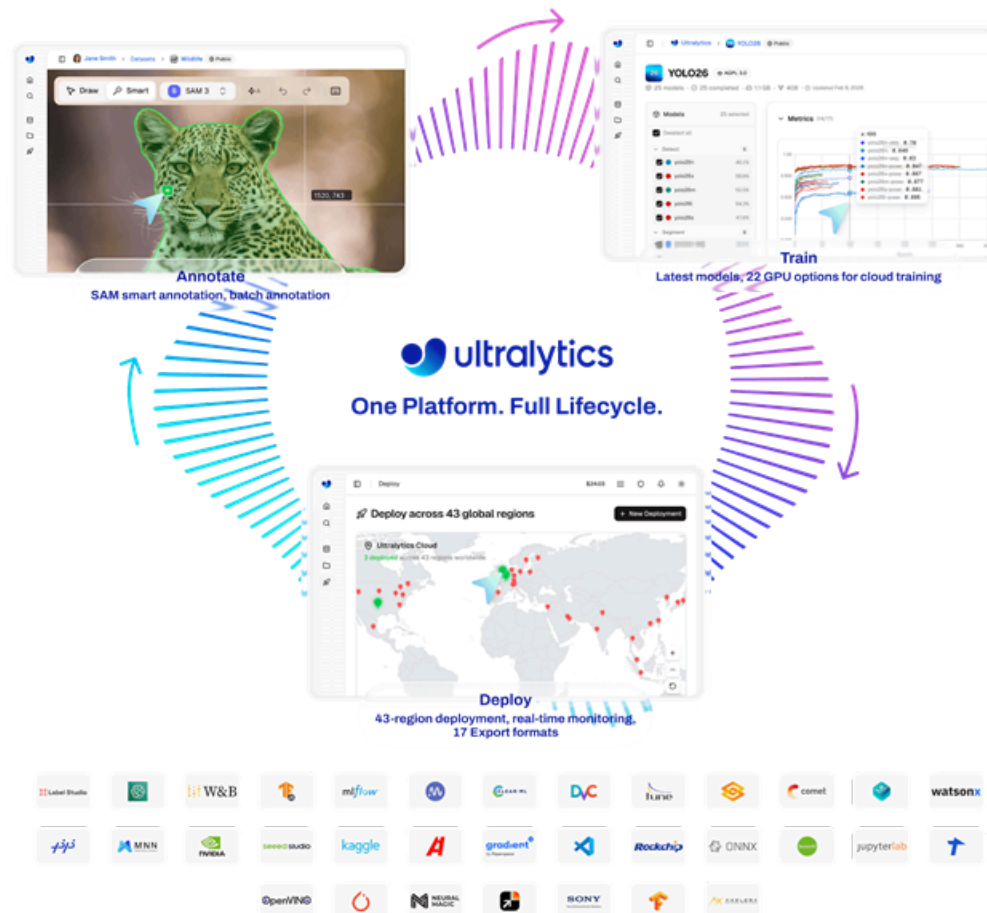


Figure 2.1 Ultralytics YOLO Computer Vision Platform (Annotate, Train, Deploy)

Source: <https://github.com/ultralytics/ultralytics>

Related Study 2: Artificial Vision for Cacao Bean Classification (Zhinin Vera et al., 2023)

Zhinin Vera et al. (2023) developed an artificial vision system to detect and classify cacao beans using a YOLOv5 model trained on an Ecuadorian dataset of *Theobroma cacao* L. beans separated into healthy and diseased classes. The proposed method achieved 88.5 percent classification accuracy and included a low-cost prototype intended to help farmers grade bean quality at the post-harvest stage.

This study is directly relevant to CacaoScan Model B because it validates YOLO-based detection on individual cacao beans rather than tray-level images. However, the system addressed only two quality classes, did not classify Philippine cultivar groups, and provided no IoT actuation or physical sorting mechanism. CacaoScan extends this line of research by combining YOLOv8 inference with ESP32-controlled bin diversion and Model A classification of Criollo, Forastero, and Trinitario beans.



Figure 2.2 Philippine Cacao Pods and Fermented Beans for Post-Harvest Quality Assessment

Source: <https://www.vancouverpcg.org/wp-content/uploads/2020/09/6-About-the-Philippine-Cacao.pdf>

Related Study 3: IoT-Integrated Citrus Sorting System in the Philippines (Nebrida, 2025)

Nebrida (2025) developed an intelligent citrus classification system using weight sensing and image processing with an Arduino microcontroller at Nueva Vizcaya State University in the Philippines. The automated system integrated load cell weight measurement with camera-based image analysis to classify citrus fruits by size and quality category, demonstrating that low-cost microcontroller platforms can execute sensor-to-actuator sorting pipelines for Philippine agricultural cooperatives.

The study is directly relevant to CacaoScan as a locally contextualized precedent for IoT-integrated physical sorting using accessible hardware. However, the system relied on rule-based sensor arrays and image thresholding rather than deep learning inference, limiting its ability to detect complex visual defects such as mold and slaty coloration on cacao surfaces.

CacaoScan advances beyond this approach by replacing rule-based sensors with dual-model YOLOv8 computer vision trained on cacao-specific defect categories.

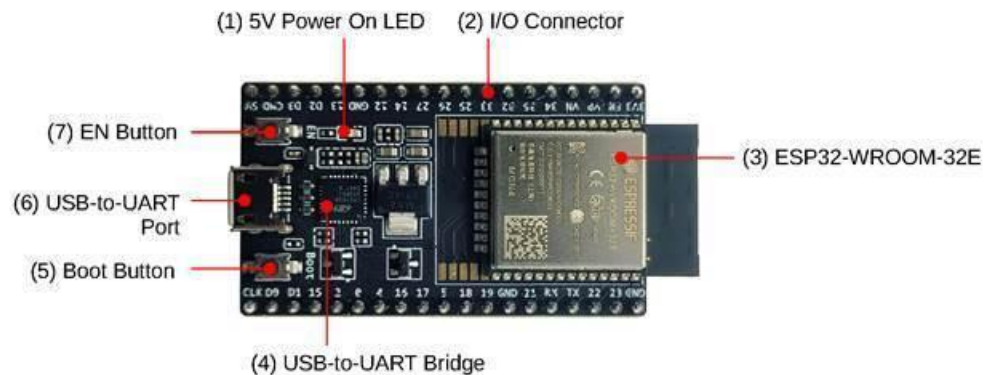


Figure 2.3 ESP32-DevKitC Development Board with ESP32-WROOM-32E Module

Source: <https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32/esp32-devkitc/index.html>

Related Study 4: Philippine Cacao Industry Roadmap (Department of Agriculture and Department of Trade and Industry, 2017)

The 2017-2022 Philippine Cacao Industry Roadmap published by the Department of Agriculture and the Department of Trade and Industry establishes the national industry context for CacaoScan. The roadmap documents that Filipino farmers cultivate three major cacao cultivar groups (Criollo, Forastero, and Trinitario) and identifies post-harvest quality consistency, export readiness, and smallholder competitiveness as strategic priorities for Mindanao and the Zamboanga Peninsula.

Although the roadmap defines the variety structure and market direction that CacaoScan Model A is designed to support, it does not prescribe an automated AIoT machine for staged cooperative-scale variety segregation and physical sorting. CacaoScan operationalizes the roadmap's cultivar framework by training Model A on Criollo, Forastero, and Trinitario classes while implementing Model B grading, three-way bin output, and mobile certification features.



Figure 2.4 Criollo, Forastero, and Trinitario Cacao Pods

Source: <https://www.vancouverpcg.org/wp-content/uploads/2020/09/6-About-the-Philippine-Cacao.pdf> (DA/DTI Philippine Cacao Industry Roadmap, 2017)

Related Study 5: Deep Computer Vision System for Cocoa Classification (Lopes et al., 2022)

Lopes, da Costa, Barbin, and Cruz-Tirado (2022) compared a Deep Computer Vision System (DCVS) using ResNet18 and ResNet50 against traditional machine learning with SVM and Random Forest for cocoa bean variety classification. Using a dataset of 1,239 samples, the best DCVS configuration achieved 96.82 percent accuracy with ResNet18, outperforming the traditional CVS at 85.71 percent with SVM.

This study is highly relevant to CacaoScan Model A because it validates deep learning for genetic variety discrimination on individual cocoa beans using region-specific samples. However, the system performed classification only without quality grading under PNS/BAFS 58:2019, physical sorting actuation, or mobile batch monitoring. CacaoScan integrates variety classification with defect-based quality grading and ESP32-controlled physical bin diversion in one cooperative-scale machine.

Related Study 6: Cocoa Bean Classification Using Enhanced Feature Extraction (Opoku Eric et al., 2023)

Opoku Eric et al. (2023) proposed enhanced image feature extraction techniques with a regularized artificial neural network for cocoa beans classification, published in Engineering Applications of Artificial Intelligence. The study demonstrated that engineered texture and color features combined with neural network classifiers can achieve strong accuracy on cocoa quality categories under controlled imaging conditions.

The work supports the feasibility of Model B quality grading for cacao but focuses on software-based classification without Philippine cultivar classes, dual-model architecture, or physical sorting integration. CacaoScan builds on this evidence by deploying YOLOv8 end-to-end detection on GPU hardware and linking inference outputs to ESP32 servo actuators for automated bin assignment.

Related Study 7: Cocoa Bean Digital Image Classification for Smart Farming (Adhitya et al., 2020)

Adhitya, Prakosa, Köppen, and Leu (2020) demonstrated a methodology for textural feature analysis on digital images of cocoa beans, comparing Gray Level Co-occurrence Matrix (GLCM) features with convolutional neural network feature extraction for classification on low-performance computational devices. Results showed that GLCM texture features could contribute reliable classification results suitable for on-site preprocessing in smart farming applications.

The study is relevant to CacaoScan's dataset strategy and image acquisition standards because it emphasizes controlled capture conditions and feature consistency for cocoa beans. However, it did not address real-time YOLOv8 inference, variety segregation into three Philippine cultivars, or automated physical sorting. CacaoScan replaces handcrafted GLCM pipelines with dual specialized YOLOv8 models and IoT actuation for cooperative post-harvest workflows.

Related Study 8: Cocoa Bean Quality Identification Using HSV and GLCM (Basri et al., 2025)

Basri et al. (2025) introduced a computer vision approach combining Hue-Saturation-Value (HSV) color features and GLCM texture features with Support Vector Machine classification to identify good-quality versus poor-quality cocoa beans captured on a conveyor-mounted acquisition system. The HSV and GLCM combination achieved up to 0.99 accuracy when adequate technical lighting was maintained during data acquisition.

This study directly supports CacaoScan's emphasis on standardized LED lighting and documented camera setup because Basri et al. showed that illumination quality strongly affects classification reliability. The system classified only two quality groups without variety identification, deep learning defect localization, or physical sorting. CacaoScan extends quality assessment to three PNS-aligned output bins plus Model A cultivar classification with ESP32 actuation.

Related Study 9: Real-Time Automatic Citrus Grading-Sorting Machine (Chakraborty et al., 2023)

Chakraborty, Subeesh, Dubey, and Jat (2023) developed an Automatic Fruit Grader (AFG) that integrates washing, vision-based accept-reject sorting, and weight grading in a single real-time machine using a custom lightweight deep convolutional neural network deployable on

Raspberry Pi edge hardware. The design targets operational throughput by combining multiple post-harvest operations that cooperatives typically perform separately.

This study is relevant to CacaoScan because it demonstrates an integrated vision-plus-physical-sorting pipeline rather than screen-only assessment. However, the AFG addresses citrus weight and color grading rather than cacao surface defects, Philippine cultivar classes, or dual-model inference. CacaoScan applies a similar sense-decide-act philosophy to individual cacao beans with YOLOv8 specialization and ESP32 servo bin control.

Related Study 10: Real-Time Fruit Grading Using Deep Learning on Raspberry Pi (Ismail and Malik, 2022)

Ismail and Malik (2022) presented a real-time visual inspection system for grading fruits using computer vision and deep learning techniques, deployed on a low-cost Raspberry Pi module with camera and touchscreen interface. The study applied and compared multiple deep learning architectures and stacking ensembles to improve grading performance under resource-constrained edge deployment.

The work validates deployable deep learning for agricultural grading accessible to smallholder facilities, aligning with CacaoScan's cloud-hosted dual-model pipeline and cooperative-scale design goals. However, the system did not perform individual bean analysis, cacao variety classification, or servo-based physical diversion. CacaoScan applies the same accessibility principle through conveyor-fed batch staging, ESP32 actuation, IR-counted vision cycles, and Supabase-backed batch logging.

Related Study 11: YOLOv8 Cherry Defect Detection for Intelligent Sorting (Liu et al., 2025)

Liu et al. (2025) proposed YOLOv8-DCPF, an enhanced YOLOv8n framework for sweet cherry surface defect and malformation detection, achieving precision of 92.6 percent, mAP of 91.2 percent, recall of 89.4 percent, and F1-score of 89.0 percent. The model integrates dilation-wise residual modules, channel attention fusion, and improved bounding box regression for small defect localization.

The reported precision, recall, and F1-score metrics align with CacaoScan's evaluation framework for Model B defect detection. However, the study addressed cherry sorting conceptually without ESP32 integration, cacao-specific defect classes, or Philippine cultivar identification. CacaoScan adopts comparable per-class metrics while extending the pipeline to dual-model inference and physical three-bin sorting.

Related Study 12: YOLOv8-Orah Mandarin Postharvest Defect Detection (Li et al., 2025)

Li et al. (2025) developed YOLOv8-Orah for postharvest surface defect detection on Orah mandarin citrus, achieving precision of 81.9 percent, recall of 78.8 percent, and average precision of 84.2 percent with reduced parameter count compared to baseline YOLOv8n. The model addresses multi-scale fruit defects and small-target detection challenges common in post-harvest grading lines.

This study reinforces the selection of YOLOv8 for post-harvest defect grading in cooperative processing environments. It does not cover cacao beans, variety classification, or automated actuator control. CacaoScan applies YOLOv8 separately to Model A and Model B with decision fusion determining Export-Grade, Needs Drying, or Rejected bin assignment under PNS/BAFS 58:2019 criteria.

Related Study 13: Multi-Scale Attention-Augmented YOLOv8 for Soybean Defect Detection (Wu et al., 2025)

Wu et al. (2025) presented a multi-scale attention-augmented YOLOv8 framework for real-time surface defect detection in fresh soybeans, integrating Squeeze-and-Excitation and Multi-Scale Dilated Attention modules to improve detection of small, low-contrast defects. Systematic ablation studies demonstrated improved precision, recall, and mAP for challenging defect categories such as wormholes and speckles.

The research supports CacaoScan's class-balancing and augmentation strategy for underrepresented mold and slaty defect classes on cacao surfaces. The soybean system remains a vision-only classifier without IoT sorting, mobile monitoring, or cacao cultivar taxonomy. CacaoScan uses independent Model B training with stratified sampling to address similar small-defect detection challenges.

Related Study 14: Real-Time Crop Sorting with ESP32-CAM and Deep Learning (Bohari et al., 2025)

Bohari et al. (2025) developed a real-time chili sorting system integrating deep learning with ESP32-CAM hardware to automate grading by ripeness and color, incorporating a motorized conveyor and microcontroller-based control for operational deployment. The YOLO-based model achieved detection accuracies exceeding 80 percent for red and green chili categories in dynamic environments.

This study is among the closest precedents to CacaoScan's ESP32 vision-and-actuation integration, validating low-cost camera modules for agricultural sorting pipelines. However, it addressed chili ripeness rather than cacao variety and PNS defect grading, and did not implement dual-model inference or digital certification logs. CacaoScan extends ESP32-CAM concepts with a dedicated inference server, dual YOLOv8 models, and Supabase real-time monitoring.

Related Study 15: Edge AI Sorting on ESP32-CAM (Mohamed, Salim, and Ibrahim, 2025)

Mohamed, Salim, and Ibrahim (2025) presented EdgeSorter, an embedded AI system deployed on ESP32-CAM using an optimized MobileNetV1 architecture that achieved 95 percent classification accuracy for three fruit types with 45 millisecond inference latency within 137.7 KB RAM constraints. The system demonstrated direct mechatronic control of sorting mechanisms from edge inference outputs.

The study validates ESP32-class hardware for decision-to-action agricultural sorting at low cost, supporting CacaoScan's IoT control layer design rationale. MobileNetV1 on ESP32-CAM lacks the defect sensitivity required for mold and slaty cacao detection compared to

GPU-trained YOLOv8 served through cloud inference endpoints. CacaoScan therefore offloads dual-model inference to Hugging Face-hosted FastAPI while using ESP32-S3 strictly for timed servo actuation and IR verification.

Related Study 16: Intelligent Banana Postharvest Sorting Using Thermal Imaging and CNN (Melesse, 2026)

Melesse (2026) introduced a lightweight CNN combined with thermal imaging for banana postharvest sorting into fresh, ripe, overripe, and rotten quality groups using 4,336 thermographic images with augmentation and class balancing. The framework reported robust accuracy with confusion matrix and precision-recall evaluation across quality thresholds.

The evaluation methodology using confusion matrices and per-class performance reporting directly aligns with CacaoScan's separate Model A and Model B assessment requirements. Thermal imaging for bananas differs from visible-surface cacao defect inspection under PNS/BAFS 58:2019, and the system did not integrate physical sorting actuators. CacaoScan applies comparable metrics to visible-spectrum dual-model grading with ESP32 bin control.

Related Study 17: Deep-Learning Wireless Visual Sensor for Mushroom Sorting (Deng et al., 2022)

Deng, Liu, and Xiao (2022) developed a deep-learning-based wireless visual sensor system for shiitake mushroom sorting using Vision Transformer training with data augmentation on small-sample datasets, achieving 99.2 percent training accuracy and 98.53 percent operational sorting accuracy with 8.7 millisecond processing time per image.

The study demonstrates that high-throughput visual sorting with stable accuracy is achievable using wireless sensor networks and augmentation strategies relevant to CacaoScan's cooperative Wi-Fi architecture. It addresses mushroom texture classes rather than cacao cultivars or export defect criteria, without mobile certification or dual-model fusion. CacaoScan adapts similar latency and accuracy targets to individual cacao bean processing.

Related Study 18: Multi-Object Robot Sorting Based on Deep Learning (Zhang et al., 2021)

Zhang et al. (2021) built a robot multi-object sorting system combining rotating target detection and Mask R-CNN instance segmentation to determine object pose, category, and grasping order in unstructured stacked scenes. Experiments demonstrated efficient, accurate, and stable sorting performance on an integrated robotic platform.

The research confirms that deep learning can drive reliable physical sort decisions when fused with actuation control logic, a principle CacaoScan applies at cooperative scale with ESP32 servo motors. Industrial robotic systems exceed the cost and complexity appropriate for barangay cacao cooperatives. CacaoScan simplifies actuation to three-bin diversion while maintaining logged inference-to-actuation synchronization.

Related Study 19: Deep Learning Fruit Detection Review (Xiao et al., 2023)

Xiao et al. (2023) published a comprehensive review of fruit detection and recognition based on deep learning for automatic harvesting, analyzing challenges including dataset scarcity, small-target detection, occlusion, multi-scale objects, and lightweight model requirements from 2018 onward. The review identifies YOLO-family detectors as leading architectures for real-time agricultural vision tasks.

As a synthesis source, this review supports CacaoScan's selection of YOLOv8 for cooperative post-harvest sorting where latency and accuracy must balance under limited hardware budgets. The review focuses on orchard harvesting rather than post-harvest bean grading, variety segregation, or IoT bin actuation. CacaoScan contributes an applied case study filling the gap between harvesting vision literature and cooperative cacao sorting machines.

Related Study 20: CNN-Based Detection in Fresh Fruit Production Review (Wang et al., 2022)

Wang et al. (2022) reviewed convolutional neural network-based detection methods across the fresh fruit production chain, including flower detection, fruit detection, harvesting, and fruit grading, documenting how improved CNN architectures address environmental variability in each production stage. The review compares data acquisition and model training practices across grading and sorting applications.

This review provides theoretical grounding for CacaoScan's dual-model training workflow, dataset augmentation, and separate evaluation of classification tasks within a single production pipeline. It does not cover cacao-specific PNS standards, Philippine cultivar taxonomy, or ESP32-integrated sorting prototypes. CacaoScan implements the review's grading-stage recommendations through Model B defect detection, Model A variety classification, and measurable F1-score thresholds.

Table 2. Review of Related Projects Comparison Table

Features	YOLOv8 Postharvest Study (Cordeiro et al., 2025)	Cacao Bean Vision System (Zhinin Vera et al., 2023)	IoT Citrus Sorter (Nebrida, 2025)	Philippine Cacao Roadmap (DA/DTI, 2017)	CacaoScan (Proposed)
Deep Learning / AI	YOLOv8	YOLOv5	Rule-Based Sensors	Policy Framework	YOLOv8 Dual-Model
Individual Bean Analysis	Yes	Yes	No	No	Yes

Philippine Cultivar Classification (Criollo, Forastero, Trinitario)	No	No	No	Defined Only	Yes
Quality Defect Detection (PNS/BAFS 58:2019)	Partial	Partial	No	Defined Only	Yes
Physical Sorting Mechanism	No	No	Yes	No	Yes
IoT Actuator Control	No	No	Arduino	No	ESP32
Mobile Monitoring App	No	No	No	No	Yes
Digital Quality Certification	No	No	No	No	Yes
Localized Context (Philippines)	No	No	Yes	Yes	Yes

Table 2 presents a feature comparison among four representative related works selected from the twenty reviewed studies because these four sources collectively cover YOLO-based defect detection, cacao vision classification, Philippine IoT sorting, and national cultivar policy. The comparison demonstrates that while each prior work validates an individual pillar of CacaoScan, none integrates dual-model YOLOv8 inference, three Philippine cultivar classes, PNS/BAFS 58:2019 defect grading, physical three-bin sorting, React Native monitoring, and Supabase-based digital certification in one affordable cooperative-scale machine.

2.3 Summary

The twenty reviewed works collectively validate the individual technical pillars of CacaoScan while confirming that none delivers the complete system proposed in this study.

Cordeiro et al. (2025), Liu et al. (2025), Li et al. (2025), and Wu et al. (2025) established YOLOv8 and enhanced YOLO variants as capable real-time architectures for post-harvest defect detection with reportable precision, recall, and F1-score. Lopes et al. (2022), Opoku Eric et al. (2023), Adhitya et al. (2020), Basri et al. (2025), and Zhinin Vera et al. (2023) demonstrated cacao and cocoa bean classification using deep learning, feature extraction, or YOLO-based vision, but without Philippine cultivar support, dual-model fusion, or physical sorting.

Nebrida (2025), Bohari et al. (2025), Mohamed et al. (2025), and Chakraborty et al. (2023) proved that IoT-integrated and edge-deployed physical sorting is feasible using accessible microcontroller and vision hardware in cooperative contexts. Melesse (2026), Deng et al. (2022), Zhang et al. (2021), Ismail and Malik (2022), Xiao et al. (2023), and Wang et al. (2022) contributed evaluation methodologies, review frameworks, and sorting system designs applicable to CacaoScan's metrics and training workflow. The Department of Agriculture and Department of Trade and Industry (2017) Philippine Cacao Industry Roadmap confirms that Criollo, Forastero, and Trinitario are the three major cultivar groups guiding national production but does not provide an automated cooperative-scale sorter.

Together, these sources confirm the research problem addressed by this study: Zamboanga cooperatives need an affordable AIoT machine that classifies each bean by Philippine cultivar group and PNS/BAFS 58:2019 quality grade, physically sorts into Export-Grade, Needs Drying, and Rejected bins, and logs results through a mobile application. No reviewed project integrates dual-model YOLOv8 inference, ESP32 actuation, and digital certification in one system designed for Barangay Ayala and Pasonanca field deployment.

2.4 Synthesis

Synthesizing the reviewed theories and twenty related studies, four design decisions for CacaoScan are justified. First, YOLOv8 dual-model inference is supported by Cordeiro et al. (2025), Liu et al. (2025), and the Ultralytics architecture (Jocher et al., 2023), aligning with Specific Objectives 2 and 3 for Model A (Criollo, Forastero, Trinitario) and Model B (Export-Grade, Needs Drying, Rejected). Second, ESP32-controlled servo sorting is supported by Nebrida (2025), Bohari et al. (2025), and Mohamed et al. (2025), which demonstrated IoT physical sorting with accessible microcontroller hardware.

Third, the dual-model YOLOv8 architecture is preferred over a single multi-class model because variety classification and PNS/BAFS 58:2019 quality grading require distinct label spaces, error costs, and dataset balancing strategies, as reflected in separate cacao variety studies (Lopes et al., 2022) and quality-focused defect detection studies (Opoku Eric et al., 2023; Basri et al., 2025). Independent training enables per-model confusion matrices with precision, recall, and F1-score reported separately.

Fourth, and most significantly, no existing system among the twenty reviewed sources integrates all proposed features, including cloud-hosted dual-model YOLOv8 inference, RTSP industrial vision, three Philippine cultivar classes, physical three-bin sorting with IR closed-loop

feedback, React Native and web dashboard monitoring, and Supabase-based digital certification, in one affordable standalone machine for Zamboanga cacao cooperatives. CacaoScan therefore occupies a unique position as the proposed solution to a documented national cultivar framework and a verified local post-harvest automation gap.

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